

MOVIE PREFERENCE ANALYSIS

ITCS544 Data Mining and Data Warehousing | Progress report draft | 28 September 2026

Executive summary

The team has integrated MovieLens 20M with Wikidata movie metadata, loaded a staged SQLite dimensional warehouse, measured five OLAP analyses, and completed an initial EDA and mining preview. The warehouse contains 20.0 million rating events. Preliminary clustering and time-split rating baselines are exploratory; final supervised model selection and tuning remain planned work.

20.0M	27,278	96.28%	5
rating events	movies	Wikidata exact-match rate	timed OLAP analyses

Project scope and sources

MovieLens supplies ratings, user-applied tags, the movie catalog, genres, and tag-genome scores. Wikidata supplies additional movie descriptions matched through normalized IMDb identifiers. The six MovieLens CSV files are related extracts from one publisher and count as one source; Wikidata is the independent second source.

Source	Role and measured coverage
MovieLens 20M	20,000,263 ratings; 465,564 source tag rows; 11,709,768 genome-score rows; 27,278 catalog movies.
Wikidata	26,263 exact matches from 27,278 usable IMDb crosswalk rows (96.28%); 981 unmatched and 34 ambiguous.

Implemented architecture

MovieLens CSVs	Wikidata metadata	Staging + audit	SQLite dimensional warehouse	OLAP + mining
Six files; raw copy preserved	Exact IMDb crosswalk; misses retained	Source rows, validation and lineage	Separate rating, tag and genome facts	Five queries; H2/H3 preview

Current implementation has a staging tier and dimensional warehouse. Separate, physically materialized data marts are not yet part of this SQLite prototype.

1. Logical and physical dimensional model

The complete relationship sketch, table definitions, SCD policy, source mappings, and bus matrix are maintained in the repository design file. This page summarizes every fact grain and its modeled relationships.

Fact grains

Fact table	One row represents	Foreign-key dimensions / measures
fact_rating	One user rating of one movie at one Unix timestamp	User, movie, UTC date, UTC time; rating; source-row lineage key.
fact_tag	One user-applied tag on one movie at one timestamp	User, movie, user tag, UTC date, UTC time; source-row lineage key.
fact_genome_score	One movie and one genome tag	Movie, genome tag; relevance from 0 to 1.

Entity relationships (ERD)

Relationship	Cardinality and join key
dim_user -> fact_rating; dim_movie -> fact_rating; dim_date -> fact_rating; dim_time -> fact_rating	One dimension row can connect to many rating facts using its surrogate key.
dim_user -> fact_tag; dim_movie -> fact_tag; dim_user_tag -> fact_tag; dim_date -> fact_tag; dim_time -> fact_tag	One dimension row can connect to many tag-event facts using its surrogate key.
dim_movie -> fact_genome_score; dim_genome_tag -> fact_genome_score	One movie or genome tag can connect to many movie/tag relevance facts.
dim_movie -> bridge_movie_genre -> dim_genre	Many-to-many movie/genre association; genre source namespace distinguishes MovieLens and Wikidata members.
dim_movie -> bridge_movie_language -> dim_language	Many-to-many association; multiple languages per movie are preserved.
dim_movie -> bridge_movie_country -> dim_country	Many-to-many association; multiple countries per movie are preserved.
dim_movie -> bridge_movie_director -> dim_person	Many-to-many association; director relationships do not multiply stored facts.

Bus matrix

Business process	User	Movie	Date/time	Genre	User tag	Genome tag	Wikidata descriptive bridges
Rating event	Yes	Yes	Yes	Via bridge	-	-	Language/country/director
User tag event	Yes	Yes	Yes	Via bridge	Yes	-	Language/country/director
Genome relevance	-	Yes	-	Via bridge	-	Yes	Language/country/director

dim_movie uses SCD Type 2 with movie_id as natural key and movie_key as version surrogate. Only the initial snapshot is loaded, so observed historical changes have not yet been measured. Rating/tag source-file plus row ordinal is a technical degenerate lineage key.

2. ETL execution and data quality

The repeatable Python and SQLite loader checks source columns, parses and stages each record with lineage, loads dimensions/bridges before facts, and records hashes, accepted/rejected/unmatched rows, and quality issues. The raw MovieLens source files remain unchanged.

Source-to-warehouse reconciliation

Input table	Before / staged	Accepted and loaded	Rejected / unresolved
Wikidata	27,278	26,263	981 unmatched; 34 ambiguous
genome-scores.csv	11,709,768	11,709,768	0 rejected; values within 0-1
ratings.csv	20,000,263	20,000,263	0 rejected; 0 outside 0.5-5.0 range
tags.csv	465,564	465,557	7 blank tag values rejected; retained in staging

Warehouse counts and checks

Post-load object	Rows	Post-load check
fact_rating	20,000,263	All source rating events loaded
fact_tag	465,557	Seven blank tag rows excluded from fact, retained in staging/issues
fact_genome_score	11,709,768	Genome relevance in [0,1]
dim_user / current dim_movie	138,493 / 27,278	One current movie row per MovieLens movie
Foreign-key violations	0	All loaded fact references resolve

Missingness and cleaning policy

No numeric rating imputation was applied. Blank user-tag text is invalid and was rejected from fact_tag while preserved in staging. Missing or ambiguous Wikidata values remain absent/null rather than being guessed. MovieLens ratings and genome scores had no out-of-range values. Missing tag/genome coverage is treated as unknown in mining, not as zero preference.

The full source hashes and row-level issue counts are in outputs/etl/etl_run_1_20260928T150023Z.json. ETL rerun and incremental-load commands are documented in docs/etl-runbook.md.

3. Warehouse OLAP benchmark

Five full-result analytical queries were run once in a fixed sequence on the 5.7 GB SQLite warehouse. Each measured time includes SQL execution and fetching every result row. The environment was macOS 26.5.1 arm64, Python 3.14.3, and SQLite 3.50.4. The database was read-only during analysis.

Query	Aggregation / OLAP operation	Result rows	Seconds
Q1	Year/month rating count and mean by MovieLens genre; drill-down and roll-up	4,792	162.935
Q2	Genre pivot by low/medium/high user activity; UTC 2010-2015 slice	20	25.890
Q3	Window ranks for highest/lowest genre means by year; >=1,000 ratings	40	81.855
Q4	Tag-event counts by user tag, genre, and year	324,427	2.268
Q5	Rating counts and means by Wikidata country and release decade	590	134.937

Observed index plans

SQLite used `ix_fact_rating_user_date` for user-history calculations and `ix_fact_rating_movie_date` for movie-level aggregation. Bridge primary-key indexes served movie-to-genre and movie-to-country lookups. Full-period Q1/Q3 scans used temporary B-trees for grouping/ranking; Q4 scanned the tag fact and used a temporary B-tree for its high-cardinality group/order.

Partitioning and benchmark limits

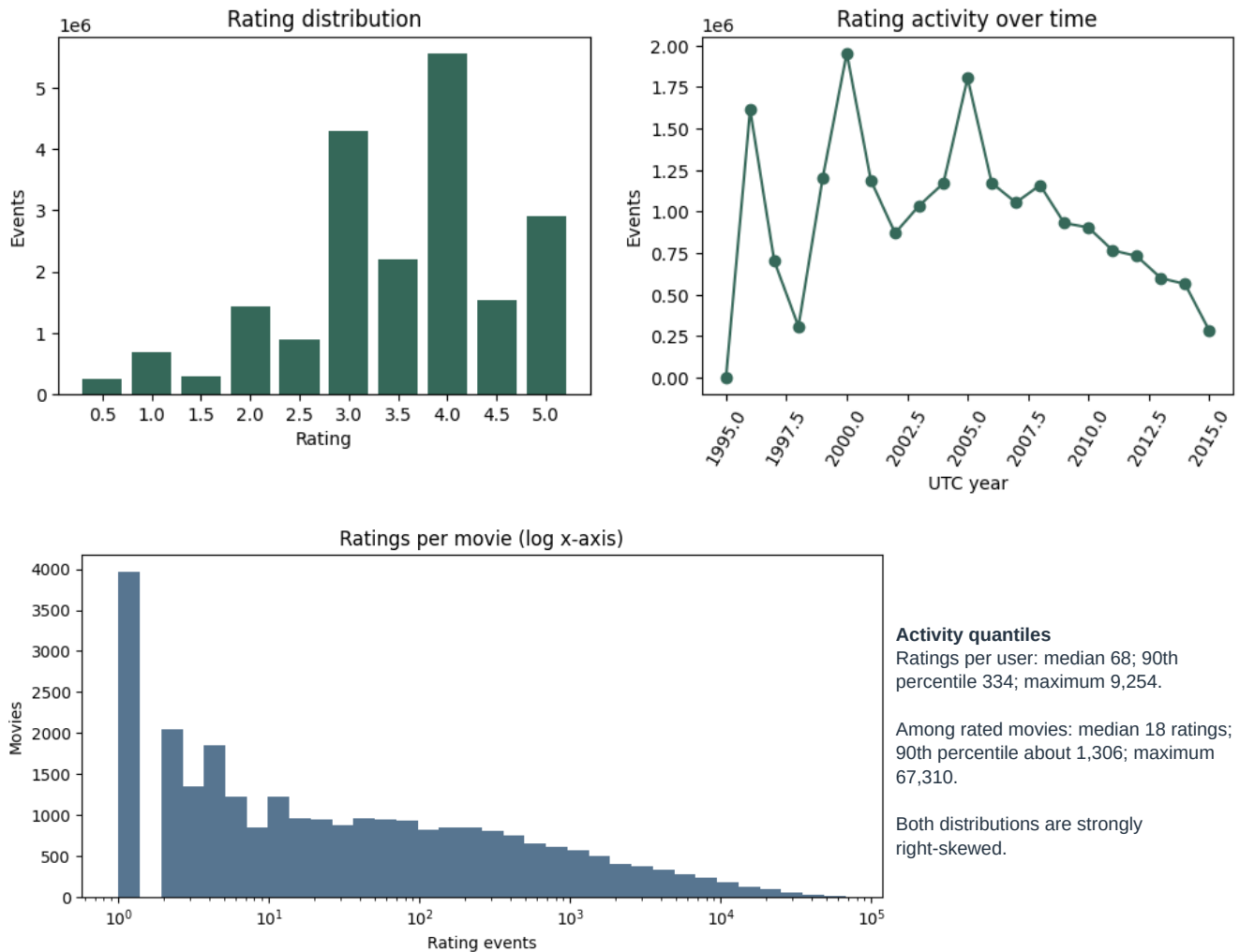
The prototype uses SQLite, which does not provide native table partitioning. No physical partitions were created; date-based partitioning would require a database engine that supports it or a different storage design. The recorded run used default SQLite and OS caches, with no warm-up or cache flush. These are observed runtimes for this machine and cache state, not universal or claimed-optimal timings.

Q5 coverage companion: 26,232 exact-match movies had country data; 31 exact matches did not; 1,015 movies had no exact Wikidata match (981 unmatched plus 34 ambiguous). The coverage query returned three groups in 0.799 seconds.

Full EXPLAIN QUERY PLAN output and methodology: [docs/olap-results.md](#) and [outputs/olap/olap_run.json](#).

4. Exploratory data analysis

The rating fact contains 20,000,263 events across 138,493 users and 27,278 catalog movies. MovieLens rating dates span 1995-01-09 through 2015-03-31 UTC. Rating 4.0 is the mode with 5,561,926 events.



Sparse-source coverage

Mining signal	Observed coverage	Interpretation
User-applied tags	19,545 movies; 7,800 users	Not available for most users/movies; absence is unknown.
Tag-genome relevance	11,709,768 movie/tag scores across 10,381 movies	Movie coverage is partial; relevance scores have no event timestamp.
Wikidata movie metadata	26,263 exact matches; 26,003 release dates	Current snapshot; excluded from H3 historical baseline.

5. Preliminary mining pipeline

Analysis code and full outputs are in notebooks/mining_preliminary.ipynb and outputs/mining/preliminary_run.json. Runtime environment: Python 3.14.3, NumPy 2.4.4, pandas 3.0.2, scikit-learn 1.8.0; seed 544.

H2: user-level clustering preview

Features: log rating count, average rating, rating standard deviation, share of ratings ≥ 4.0 , and genre-associated rating-event shares over 20 MovieLens genre members. The 24 features were standardized. Candidate k values were compared on a seeded 25,000-user sample; silhouette used 1,500 sampled users. The selected $k=2$ was fitted to all 138,493 users; its silhouette was estimated on 2,000 users.

k	Sampled silhouette
2	0.1170
3	0.1148
4	0.0896

Cluster	Users	Median events	Mean rating	Mean within-user SD	≥ 4 share	Top genre shares
0	49,938	65	3.593	0.968	53.6%	Comedy, Drama, Adventure
1	88,555	69	3.647	0.930	55.7%	Drama, Comedy, Thriller

Full-fit sampled silhouette: 0.1104. This suggests weak cluster separation; the two profiles have similar rating averages. The clustering is descriptive and does not confirm the proposed preference archetypes.

H3: chronological rating baselines

Train cutoff: before 2010-01-01 UTC. Test: on/after the cutoff. Training statistics are computed using only pre-cutoff events. Test movies without a training mean fall back to the training global mean.

Baseline	Test events	RMSE	MAE	Cold-start fallback events
training global mean	3,847,547	1.0194	0.7899	0
training movie mean (global fallback)	3,847,547	0.9460	0.7189	588,406

Training events: 16,152,716; held-out events: 3,847,547. The movie-average baseline performed better than the global baseline on this split. No supervised feature model has been evaluated yet, so H3's improvement hypothesis remains untested.

6. Limitations and next steps

The progress report establishes the warehouse and a repeatable preliminary mining path. The following limits should remain visible in the presentation and report:

Topic	Current evidence and limitation
Source coverage	Wikidata includes unmatched/ambiguous records; tags and genome scores have partial coverage. Missing external metadata is retained as missing.
SCD Type 2	Code supports observed changes across snapshots, but only one initial metadata snapshot has been loaded.
OLAP benchmark	Measured on one 5.7 GB SQLite database and one machine. Query timings include full fetch and depend on cache state. SQLite has no native partitioning.
H2 clustering	Silhouette around 0.11 indicates weak separation for current standardized features. Feature weighting and interpretation need later review.
H3 prediction	Only global and movie-average baselines have been measured. Final supervised model, feature ablations, and model-versus-baseline comparison are outstanding.
Temporal alignment	All H3 baseline statistics are training-only. Genome scores have no event timestamps; current Wikidata metadata was excluded from the historical baseline.

Work planned for the later mining milestone

Complete the H2 feature and cluster evaluation, fit and evaluate an H3 supervised model on the same chronological holdout, compare RMSE/MAE with these baselines, and test tag/genome feature coverage and temporal limitations. TASK-004 tracks this later work. The progress-report draft should be checked by the team, updated with member details if required, and formatted to the course eight-page maximum.

References

1. GroupLens. MovieLens 20M README and usage terms. <https://files.grouplens.org/datasets/movielens/ml-20m-README.html>
2. Wikidata. Data access and reuse. <https://www.wikidata.org/wiki/Wikidata:Reuse>
3. ITCS544 course project requirements. See docs/course-requirements.md and the supplied 2026T1_L03p_ITCS544_Project.pdf.

Reproducibility files

Warehouse design: docs/warehouse-design.md. ETL runbook and run record: docs/etl-runbook.md and outputs/etl/etl_run_1_20260928T150023Z.json. OLAP notebook/results: notebooks/olap_analysis.ipynb and docs/olap-results.md. Mining notebook/results: notebooks/mining_preliminary.ipynb and docs/mining-preliminary-results.md.